

MARK SCHEME for the October/November 2007 question paper

9709 MATHEMATICS

9709/07

Paper 7, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2007 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

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1 $H_0 p = 0.3$ $H_1 p < 0.3$ $P(0, 1, 2) = 0.7^{18} + 0.3 \times 0.7^{17} \times {}_{18}C_1$ $+ 0.3^2 \times 0.7^{16} \times {}_{18}C_2$ $= 0.001628 + 0.01256 + 0.04576$ $= 0.0599$ This is > 0.05 Accept Isaac's claim. OR Using $N(0.3, 0.0116)$ $H_0 p = 0.3$ $H_1 p < 0.3$ $z = \frac{0.111 + 1/36 - 0.3}{\sqrt{0.0116}} = -1.49159$ $-1.49159 > -1.645$ Accept Isaac's claim OR Using $N(5.4, 3.78)$ $H_0 \mu = 5.4$ $H_1 \mu < 5.4$ $z = \frac{2.5 - 5.4}{\sqrt{3.78}} = -1.49159$ $-1.49159 > -1.645$ Accept Isaac's claim	B1	Both hypotheses correct	
	M1	For finding $P(0, 1, 2)$ at least two terms of this sum needed	
	A1	Correct answer accept 0.06(0)	
	M1	Comparing with 0.05 must be 0.05	
	A1ft	Correct conclusion ft their test statistic – no contradictions	
	B1	Both hypotheses correct	
	M1	For attempt at z with or without cc	
	A1	For correct z	
	M1	For comparison	
	A1ft	Correct conclusion ft their test statistic	
	B1	Both hypotheses correct	
	M1	For attempt at z with or without cc	
	A1	For correct z	
	M1	For comparison	
	A1ft	Correct conclusion ft their test statistic	5
2 (i) $-1.645 = \frac{c - 3.2}{1.4 / \sqrt{10}}$ $c = 2.47$ rejection region is $\bar{x} < 2.47$ (ii) $m < 2.47$	M1 B1 A1 A1ft	For standardising, must have sq rt. and z value For ± 1.645 used For 2.47 For inequality correct way round (ft their 2.47 but must be < 3.2)	4
	B1ft	ft on their (i)	1
3 (i) a sample where every element has an equal chance of being chosen OR a random sample of size n is a sample chosen in such a way that each possible group of size n has the same chance of being picked.	B1		1
(ii) 130/350 (0.371) $0.371 \pm 1.96 \times \sqrt{\frac{(0.371)(0.629)}{350}}$ $= 0.371 \pm 0.050609$ $= (0.321, 0.422)$	B1 M1 B1 A1	For proportion used Correct shape $\bar{x} \pm zs / \sqrt{n}$ Correct z value 1.96 used Correct limits (written as interval)	4
(iii) $1.96 \sqrt{\frac{(0.371)(0.629)}{n}} = 0.02$ $n = 2241 \text{ or } 2242 \text{ or } 2243 \text{ or } 2240$	M1* M1*dep A1	Seeing an equation involving 0.02 or 0.04, n in denom and a sq rt and proportions used For equation of correct form Correct whole number answer	3

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4 (i) $E(\text{cost to Stella}) = 600 + 5.52 \times 500$ $= 3360$ $\text{Var}(\text{cost to Stella}) = 5.52^2 \times 7.1^2$ $= 1540 (1536)$	M1 A1 M1 M1 A1	For multiplying by 5.52 and adding 600 Correct mean For mult $7.1/7.1^2/50.41^2$ by 5.52² For $5.52^{(2)} \times 7.1^{(2)}$ or 50.41^2 with no addition/subtraction For correct answer
(ii) $P(D > 2S) = P(D - 2S > 0)$ $D - 2S \sim N(-120, 421 + 4 \times 1536)$ $\sim N(-120, 6565)$ $P(D - 2S > 0) = P\left(z > \frac{120}{\sqrt{6565}}\right)$ $= P(z > 1.481)$ $= 0.0693$	M1 B1 A1ft M1 A1	For attempt $(D - 2S)$ (or equiv) either $<$ or > 0 For correct mean (seen or implied) For correct unsimplified variance For standardising attempt For correct answer, accept 0.069
5 (i) $E(X) = \int_0^b \frac{x}{b} dx = \left[\frac{x^2}{2b} \right]_0^b = \frac{b}{2}$ $\text{Var}(X) = \int_0^b \frac{x^2}{b} - \frac{b^2}{4} = \frac{b^2}{12}$	B1 M1 A1	Correct answer (accept unsimplified) For (substituted) attempt at $\int x^2 f(x) dx - [E(X)^2]$ ie $-[E(X^2)]$ must be seen even if ignored in next line Correct answer. Accept unsimplified – but must be a single fraction.
(ii) $9.5 = b/2$ $b = 19$ AG	M1 A1	Equating their mean to their 9.5 Correct answer
(iii) 8/19 or 0.421	B1	Correct answer
(iv) $\bar{X} \sim N(9.5, 30.08/336)$ or using totals $N(3192, 10106.88)$ $P(\bar{X} < 9) = P\left(z < \frac{9 - 9.5}{\sqrt{30.08/336}}\right)$ or equiv $= P(z < -1.671)$ $= 1 - 0.9526$ $= 0.0474$	M1 A1ft M1 A1	Dividing their $b^2/12$ by 336 Correct mean and variance Standardising (must involve 336) and area < 0.5 or consistent with their figures Correct answer

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6 (i) $\frac{e^{-\lambda} \lambda^2}{2!} = 3 \frac{e^{-\lambda} \lambda^4}{4!}$ $\lambda = 2$ new $\lambda = 7$ $P(X > 3) = 1 - e^{-7} \left(1 + 7 + \frac{7^2}{2!} + \frac{7^3}{3!} \right)$ $= 0.918$	M1	Poisson equation involving λ
	A1	Correct mean
	B1ft	New mean ft $3.5 \times$ previous one
	M1	Poisson probs with their mean (at least 3 probs) and 1-
(ii) (a) $\lambda = 1.3k$ $P(X > 0) = 1 - e^{-1.3k} = 0.96$ $0.04 = e^{-1.3k}$ $k = 2.48$	A1	5 Correct answer
	B1	Correct new mean
	M1	Equation with k or λ in involving $1 - P(0) = 0.96$
	A1	correct equation
(b) $X \sim N(1300, 1300)$ $P(X > 1250) = P\left(z > \frac{1250.5 - 1300}{\sqrt{1300}}\right)$ $= P(z > -1.373)$ $= 0.915$	A1	4 correct answer
	B1	Correct new mean
	M1	Equation with k or λ in involving $1 - P(0) = 0.96$
	A1	correct equation
(b) $X \sim N(1300, 1300)$ $P(X > 1250) = P\left(z > \frac{1250.5 - 1300}{\sqrt{1300}}\right)$ $= P(z > -1.373)$ $= 0.915$	A1	3 correct answer
	B1	correct mean and variance
	M1	standardising must have sq rt with or without cc
	A1	correct answer